

INTRODUCTION TO CLOUD COMPUTING

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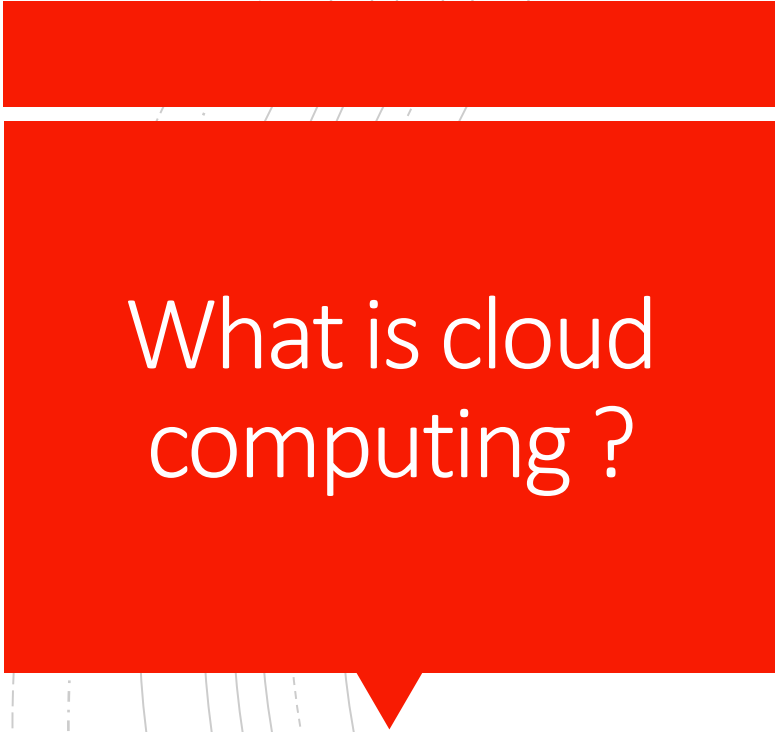
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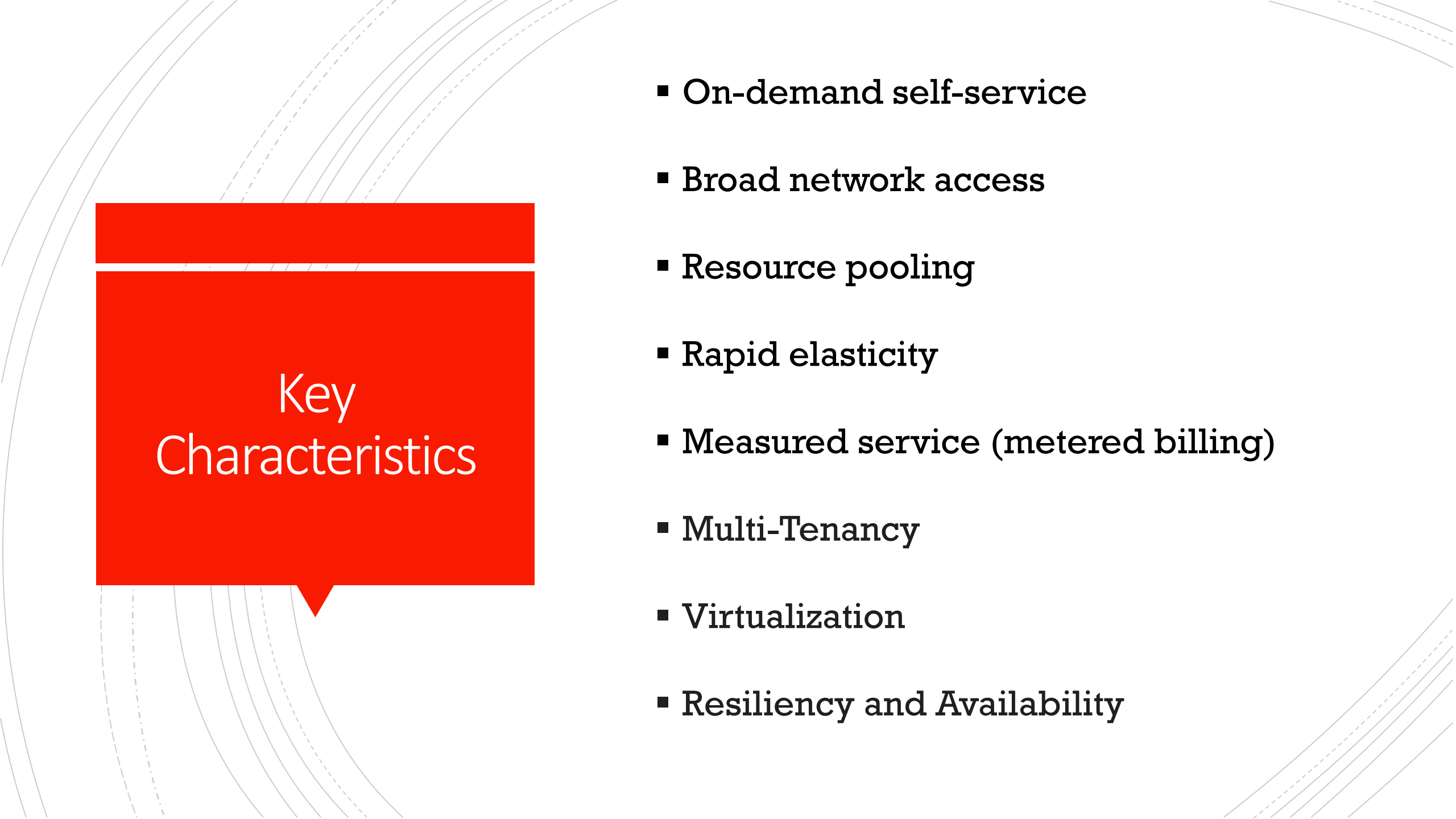
Why Learn Cloud Computing?

- **High Industry Demand**
- **Cost Efficiency**
- **Scalability & Flexibility**
- **Faster Development & Deployment**
- **Global Access & Collaboration**
- **Reliability & Disaster Recovery**
- **Supports Modern Technologies**
- **Career Growth & Future Proof Skills**

A red speech bubble with a white question mark inside, pointing towards the definition of cloud computing.

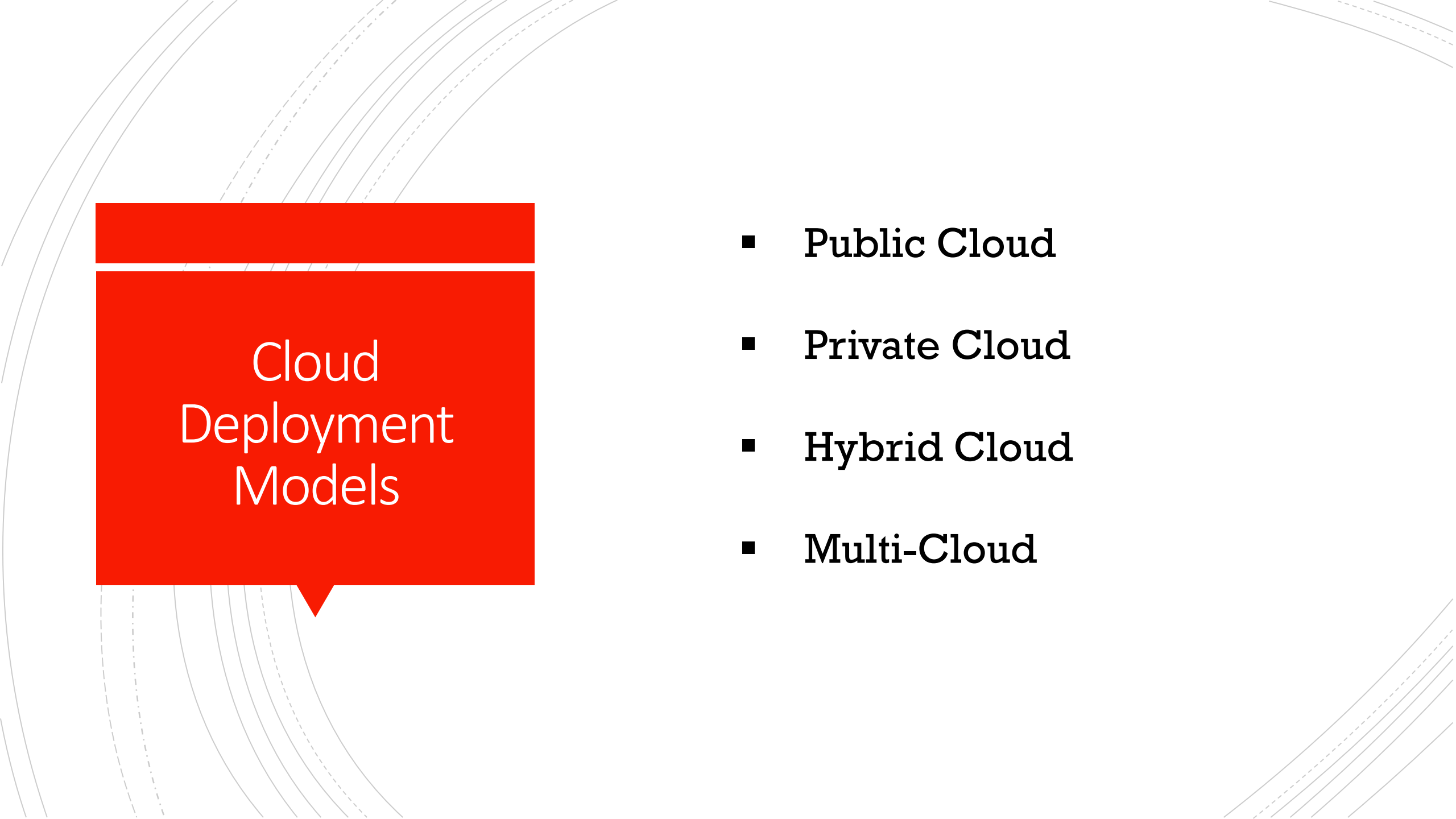
What is cloud computing ?

Cloud computing is the on-demand delivery of IT resources over the Internet with pay-as-you-go pricing.

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Key Characteristics

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service (metered billing)
- Multi-Tenancy
- Virtualization
- Resiliency and Availability

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Cloud Deployment Models

- **Public Cloud**
- **Private Cloud**
- **Hybrid Cloud**
- **Multi-Cloud**

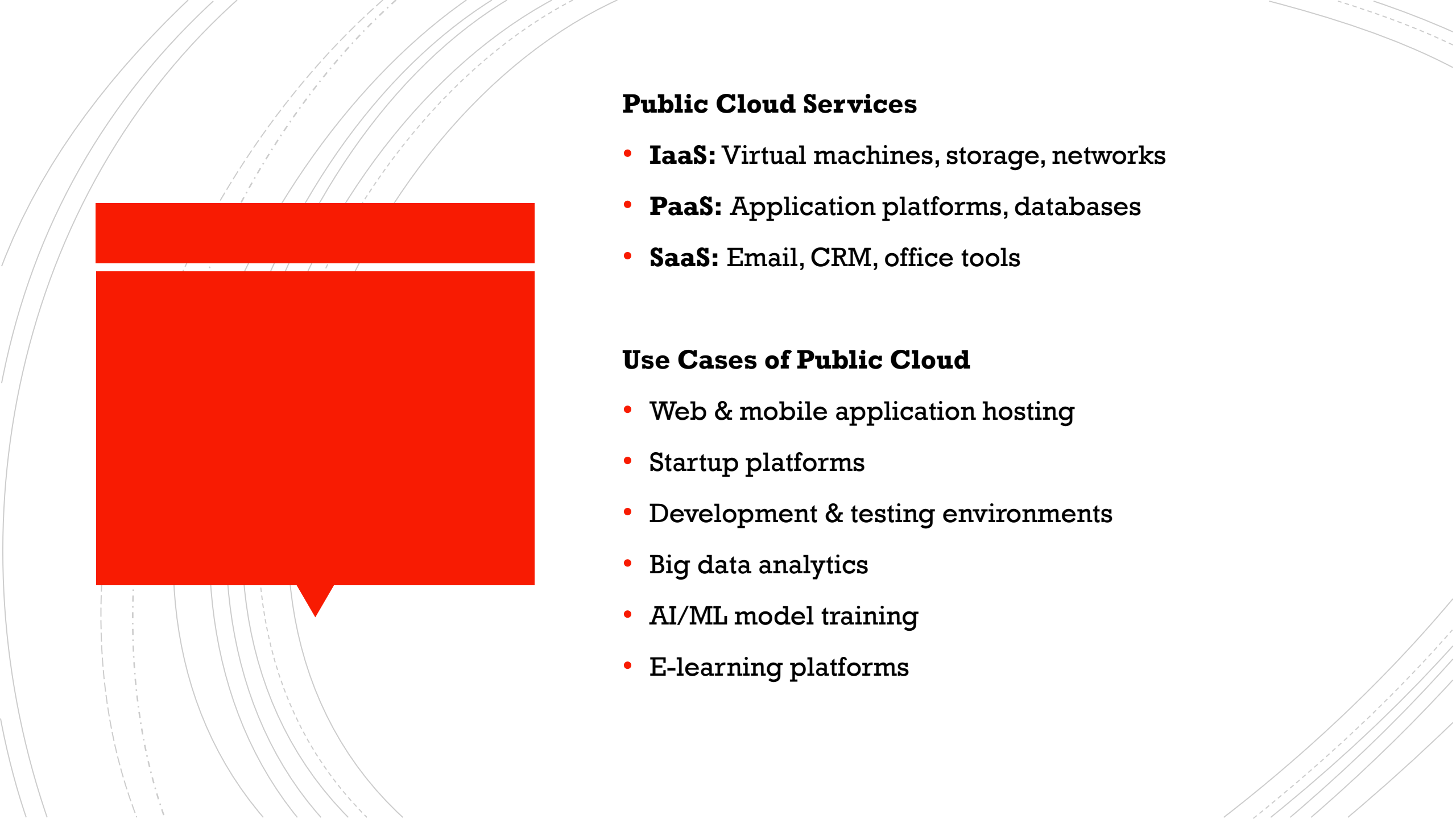
Public Cloud

A **Public Cloud** is a cloud computing model where **services are provided over the internet** by a third-party cloud provider and **shared among multiple customers**.

Infrastructure (servers, storage, networking) is **owned and managed by the cloud provider**.

Key Characteristics

- Shared infrastructure (multi-tenant)
- Pay-as-you-go pricing
- High scalability and elasticity
- Managed and maintained by provider
- Accessible from anywhere via internet

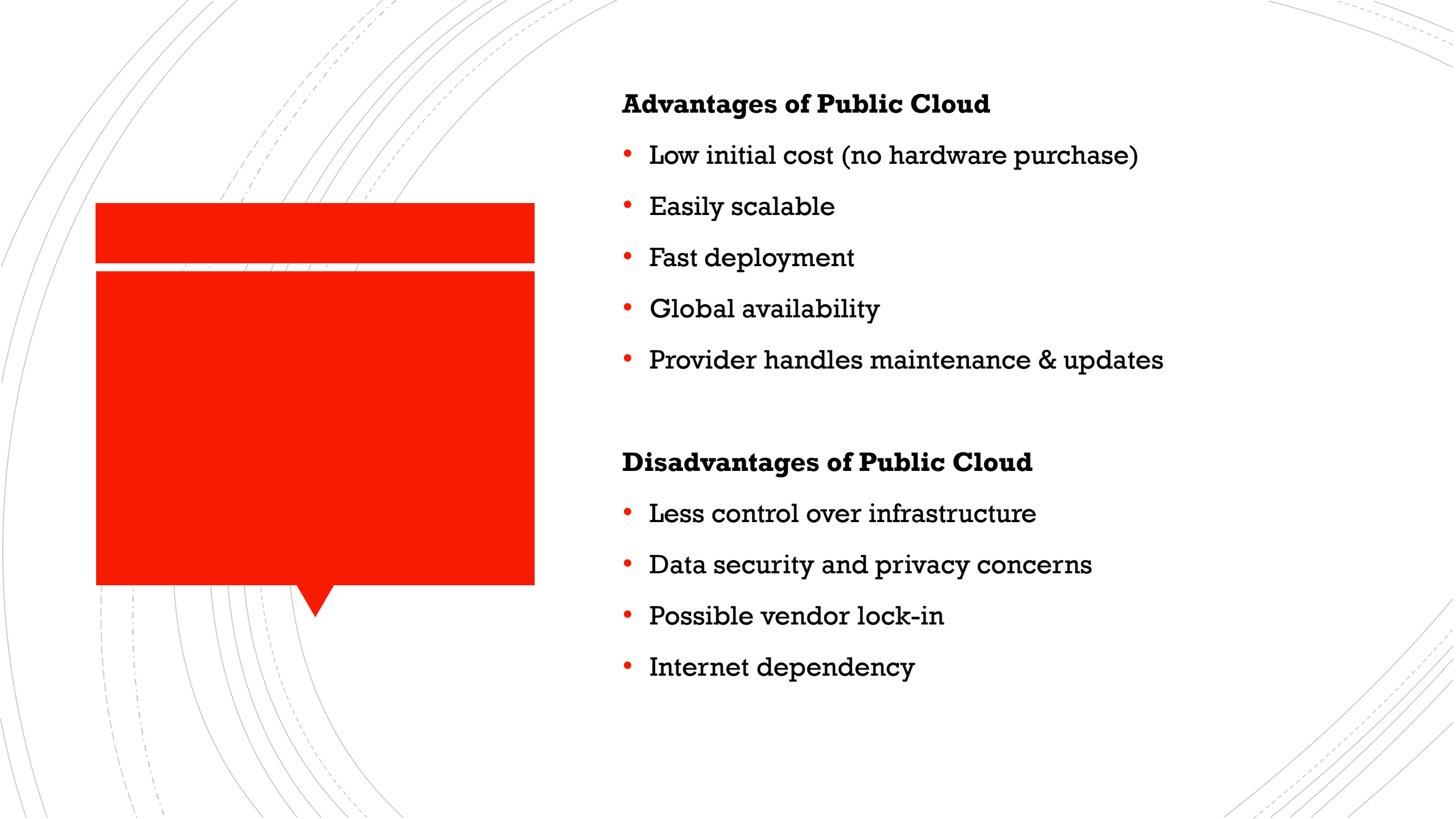


Public Cloud Services

- **IaaS:** Virtual machines, storage, networks
- **PaaS:** Application platforms, databases
- **SaaS:** Email, CRM, office tools

Use Cases of Public Cloud

- Web & mobile application hosting
- Startup platforms
- Development & testing environments
- Big data analytics
- AI/ML model training
- E-learning platforms

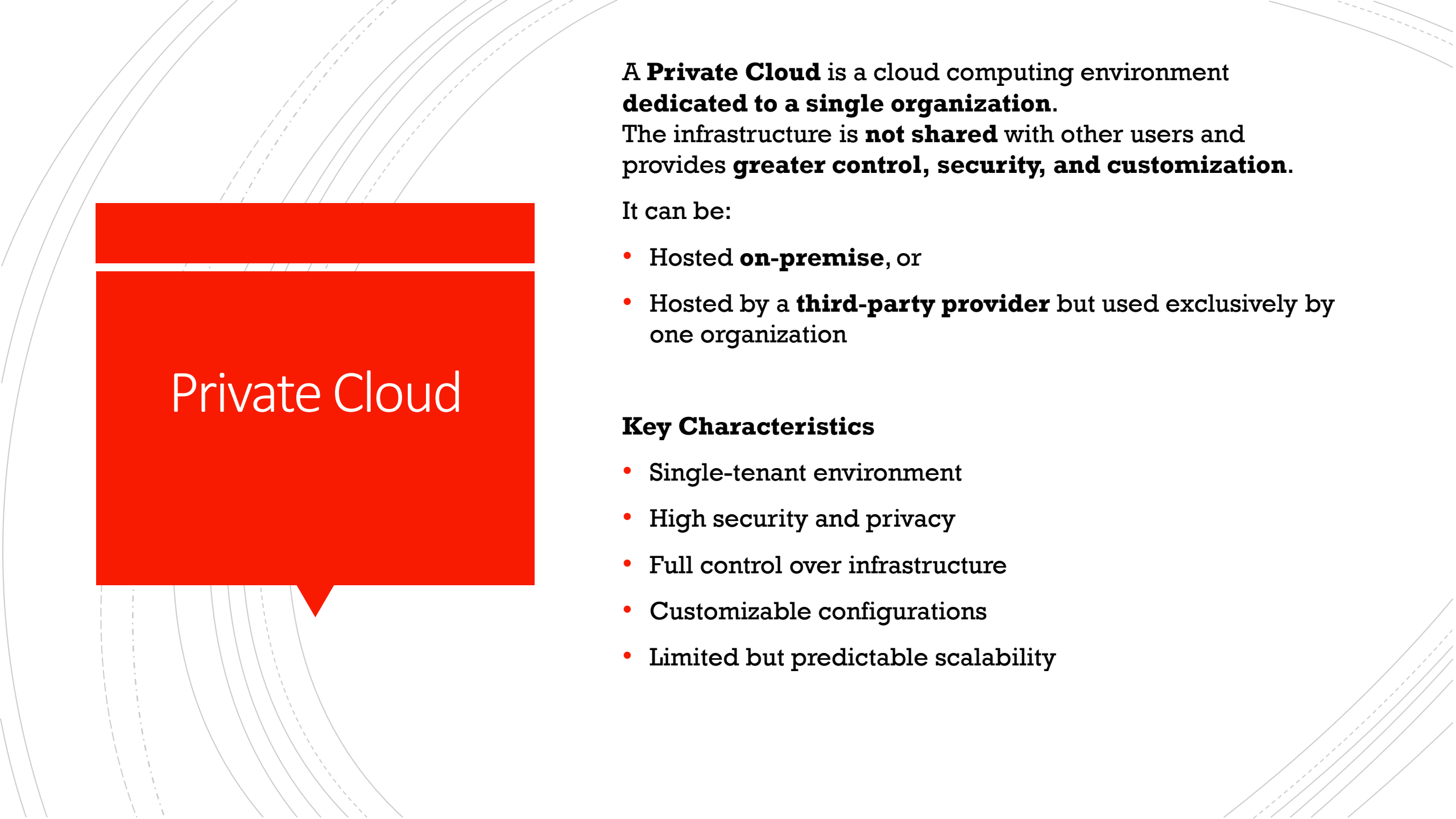


Advantages of Public Cloud

- Low initial cost (no hardware purchase)
- Easily scalable
- Fast deployment
- Global availability
- Provider handles maintenance & updates

Disadvantages of Public Cloud

- Less control over infrastructure
- Data security and privacy concerns
- Possible vendor lock-in
- Internet dependency

A decorative background featuring several concentric, curved lines in light gray and white, creating a sense of depth and movement. On the left side, there is a large red speech bubble with a tail pointing towards the bottom center. The text 'Private Cloud' is written in white inside this bubble.

Private Cloud

A **Private Cloud** is a cloud computing environment **dedicated to a single organization**.

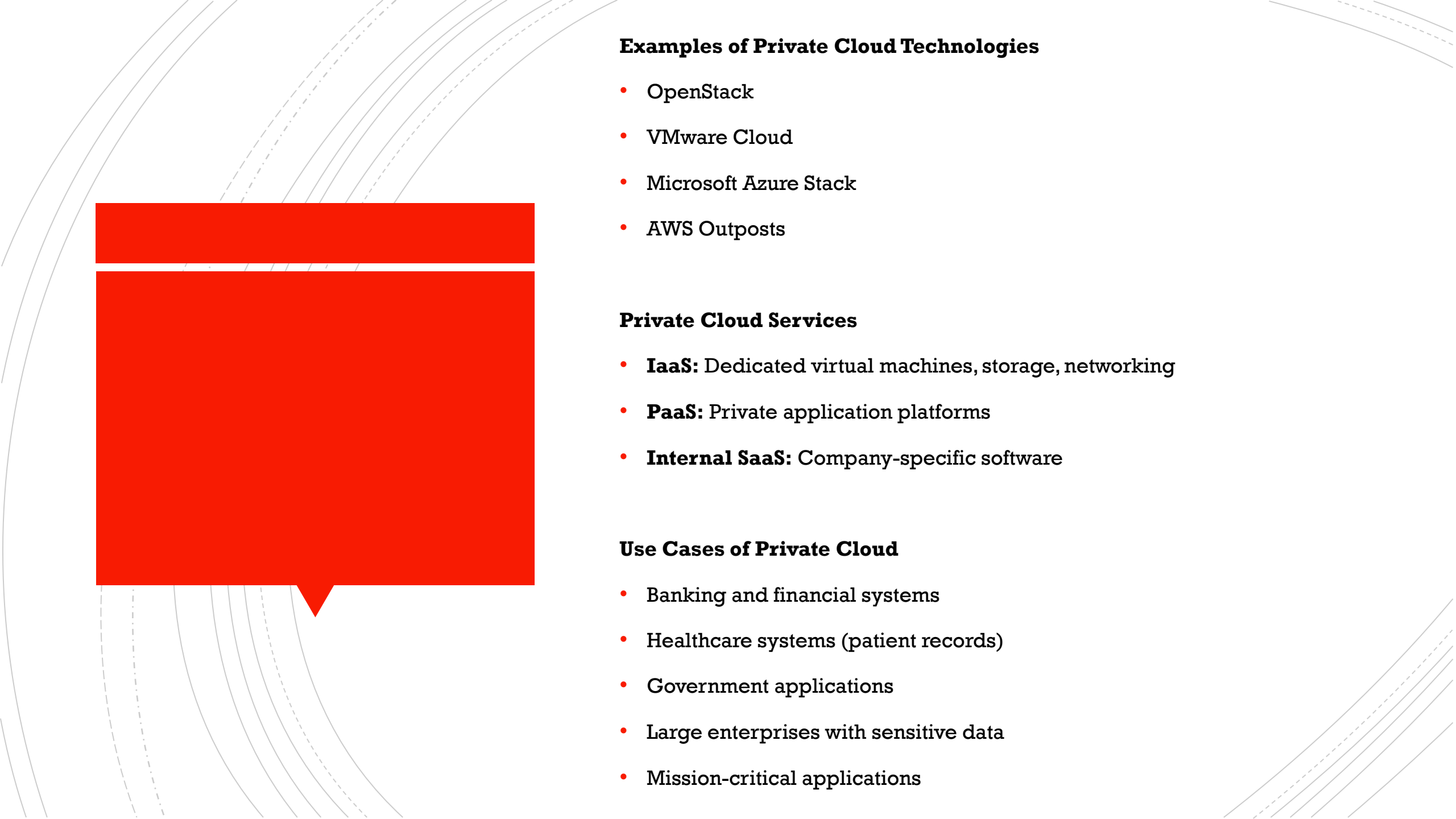
The infrastructure is **not shared** with other users and provides **greater control, security, and customization**.

It can be:

- Hosted **on-premise**, or
- Hosted by a **third-party provider** but used exclusively by one organization

Key Characteristics

- Single-tenant environment
- High security and privacy
- Full control over infrastructure
- Customizable configurations
- Limited but predictable scalability



Examples of Private Cloud Technologies

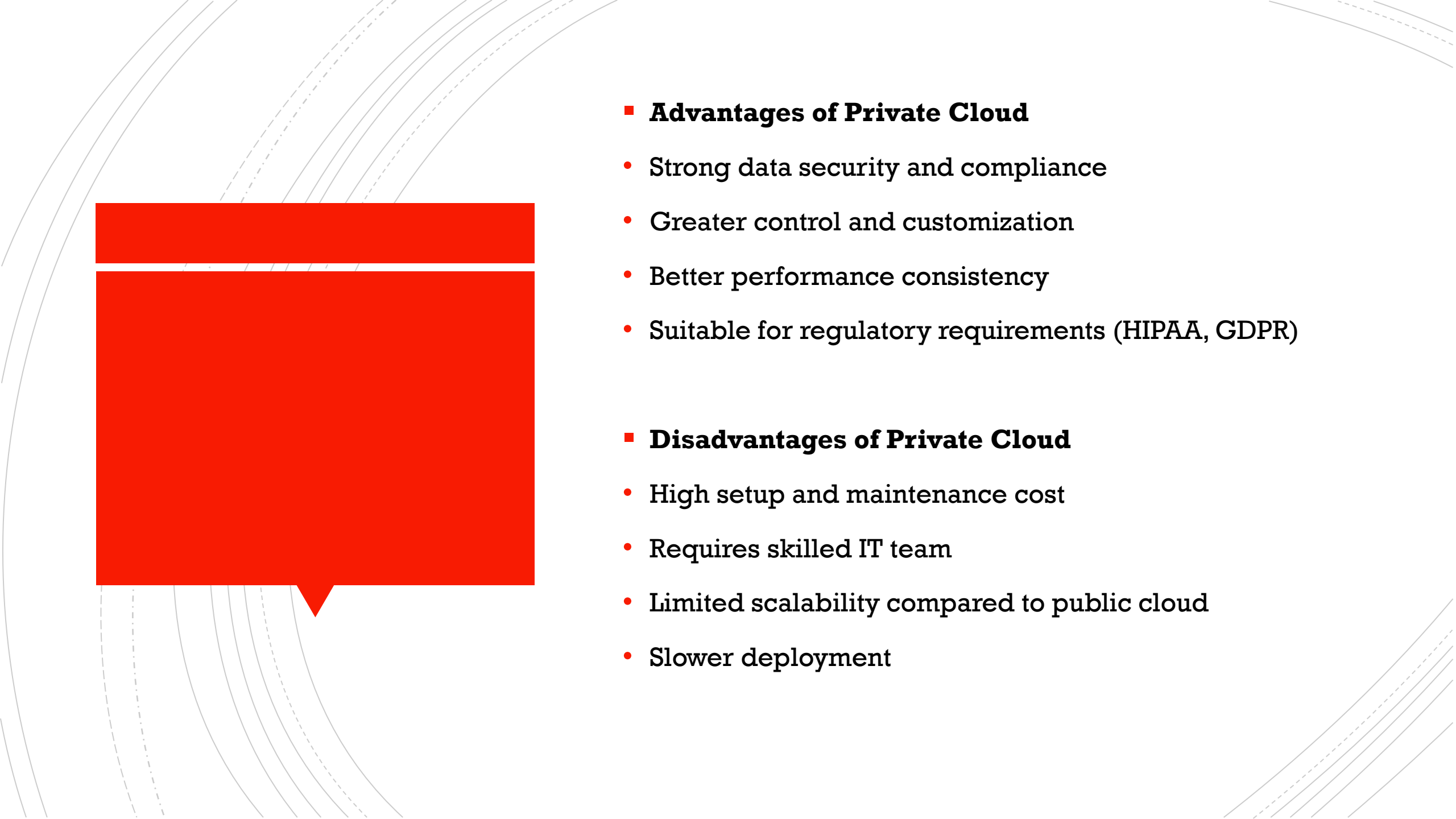
- OpenStack
- VMware Cloud
- Microsoft Azure Stack
- AWS Outposts

Private Cloud Services

- **IaaS:** Dedicated virtual machines, storage, networking
- **PaaS:** Private application platforms
- **Internal SaaS:** Company-specific software

Use Cases of Private Cloud

- Banking and financial systems
- Healthcare systems (patient records)
- Government applications
- Large enterprises with sensitive data
- Mission-critical applications



■ **Advantages of Private Cloud**

- Strong data security and compliance
- Greater control and customization
- Better performance consistency
- Suitable for regulatory requirements (HIPAA, GDPR)

■ **Disadvantages of Private Cloud**

- High setup and maintenance cost
- Requires skilled IT team
- Limited scalability compared to public cloud
- Slower deployment

Hybrid Cloud

- **What Is a Hybrid Cloud?**
- A **Hybrid Cloud** combines **two or more different environments**:
 - **Private Cloud / On-Premise infrastructure**
 - **Public Cloud**
- These environments are **integrated and work together**, allowing data and applications to move between them.

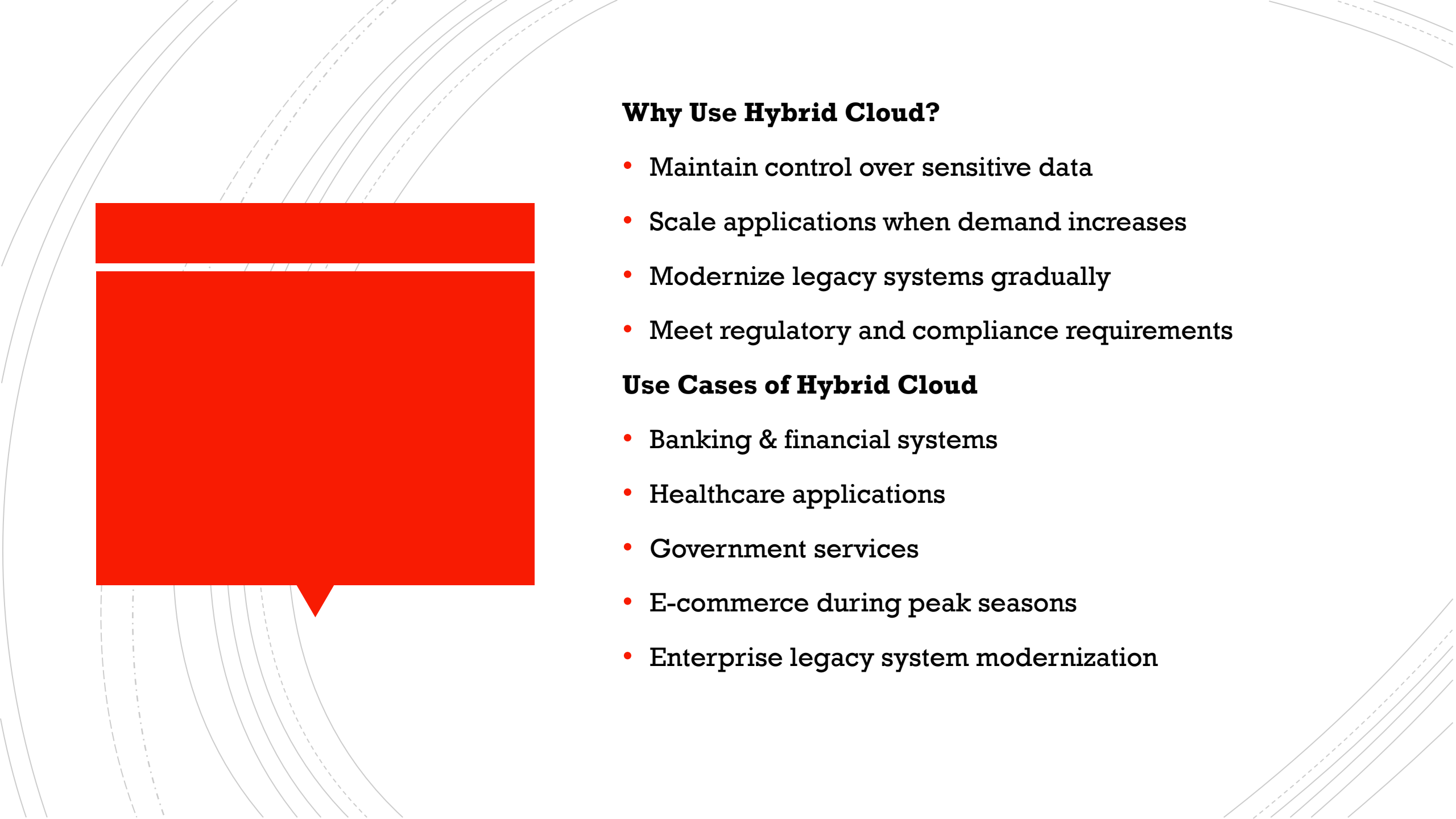


Key Characteristics

- Mix of public and private cloud
- Secure data stays on private cloud
- Public cloud used for scalability
- Workload portability
- Centralized management

Hybrid Cloud Architecture

- Sensitive data → **Private Cloud**
- High traffic or compute-intensive tasks → **Public Cloud**
- Secure connection between environments (VPN / Direct Connect)

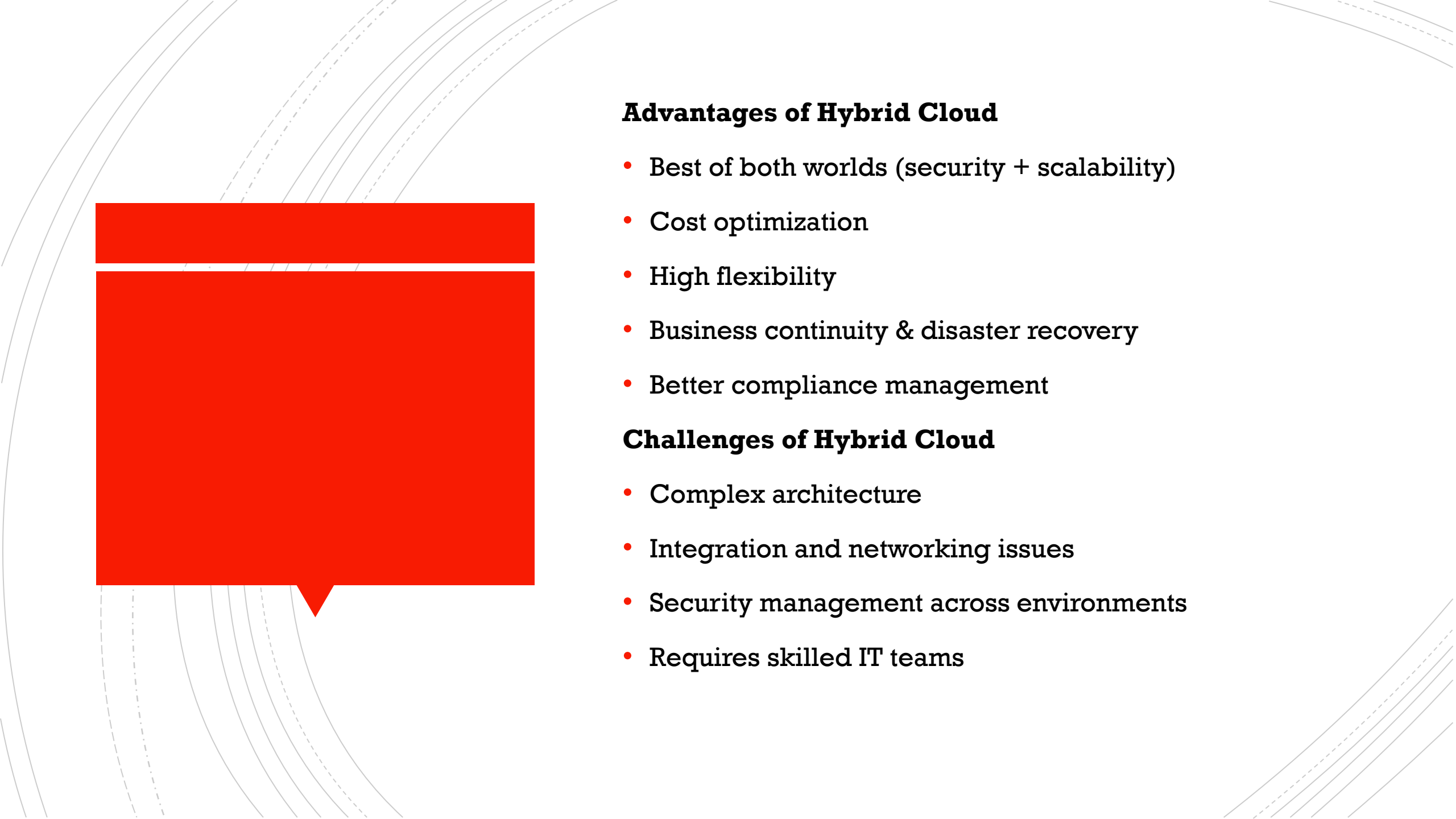
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Why Use Hybrid Cloud?

- Maintain control over sensitive data
- Scale applications when demand increases
- Modernize legacy systems gradually
- Meet regulatory and compliance requirements

Use Cases of Hybrid Cloud

- Banking & financial systems
- Healthcare applications
- Government services
- E-commerce during peak seasons
- Enterprise legacy system modernization



Advantages of Hybrid Cloud

- Best of both worlds (security + scalability)
- Cost optimization
- High flexibility
- Business continuity & disaster recovery
- Better compliance management

Challenges of Hybrid Cloud

- Complex architecture
- Integration and networking issues
- Security management across environments
- Requires skilled IT teams



Technologies Used in Hybrid Cloud

- Kubernetes
- VPN / Direct Connect
- Terraform
- Identity & Access Management (IAM)
- Monitoring & logging tools



■ **Real-World Example**

■ Banking System

- Customer data stored in private cloud
- Public cloud used for analytics & mobile apps
- Secure integration between both

■ **When to Use Hybrid Cloud?**

- When data privacy is critical
- When gradual cloud migration is needed
- When regulatory compliance is required

Multi-Cloud

What Is Multi-Cloud?

- **Multi-Cloud** is a cloud strategy where an organization uses **services from multiple cloud providers** at the same time.
- Example:
Using **AWS for compute**, **Azure for enterprise apps**, and **Google Cloud for analytics**.

Key Characteristics

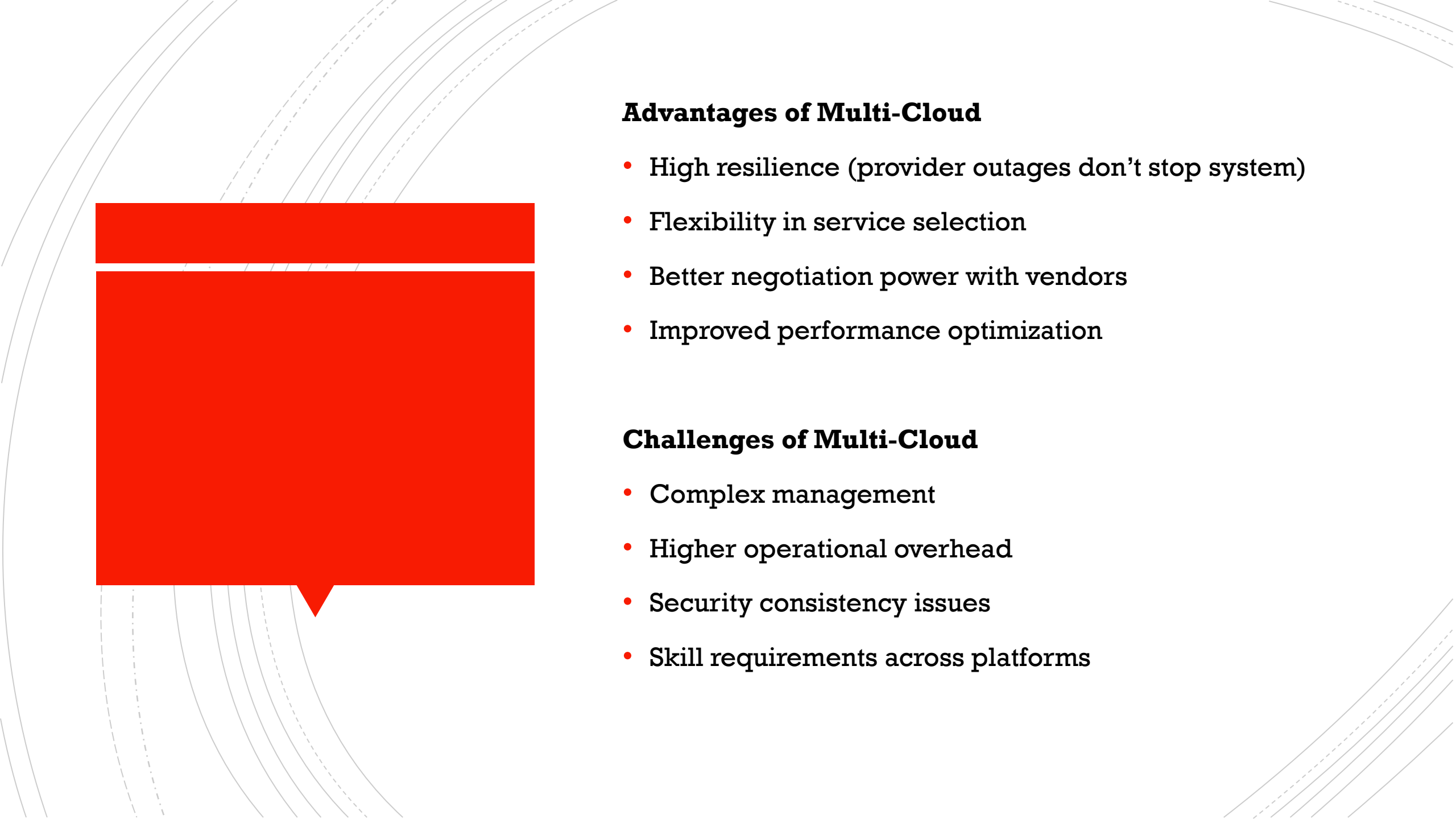
- Uses more than one cloud provider
- Can include public and private clouds
- Avoids dependency on a single vendor
- Each cloud used for its strengths

Why Use Multi-Cloud?

- Avoid vendor lock-in
- Improve reliability & availability
- Optimize cost
- Use best-of-breed services
- Meet regional or legal requirements

Use Cases of Multi-Cloud

- Global applications with regional compliance
- Disaster recovery across providers
- Large enterprises with diverse workloads
- High-availability systems
- AI/ML workloads using specialized services

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Advantages of Multi-Cloud

- High resilience (provider outages don't stop system)
- Flexibility in service selection
- Better negotiation power with vendors
- Improved performance optimization

Challenges of Multi-Cloud

- Complex management
- Higher operational overhead
- Security consistency issues
- Skill requirements across platforms

Technologies Used in Multi-Cloud

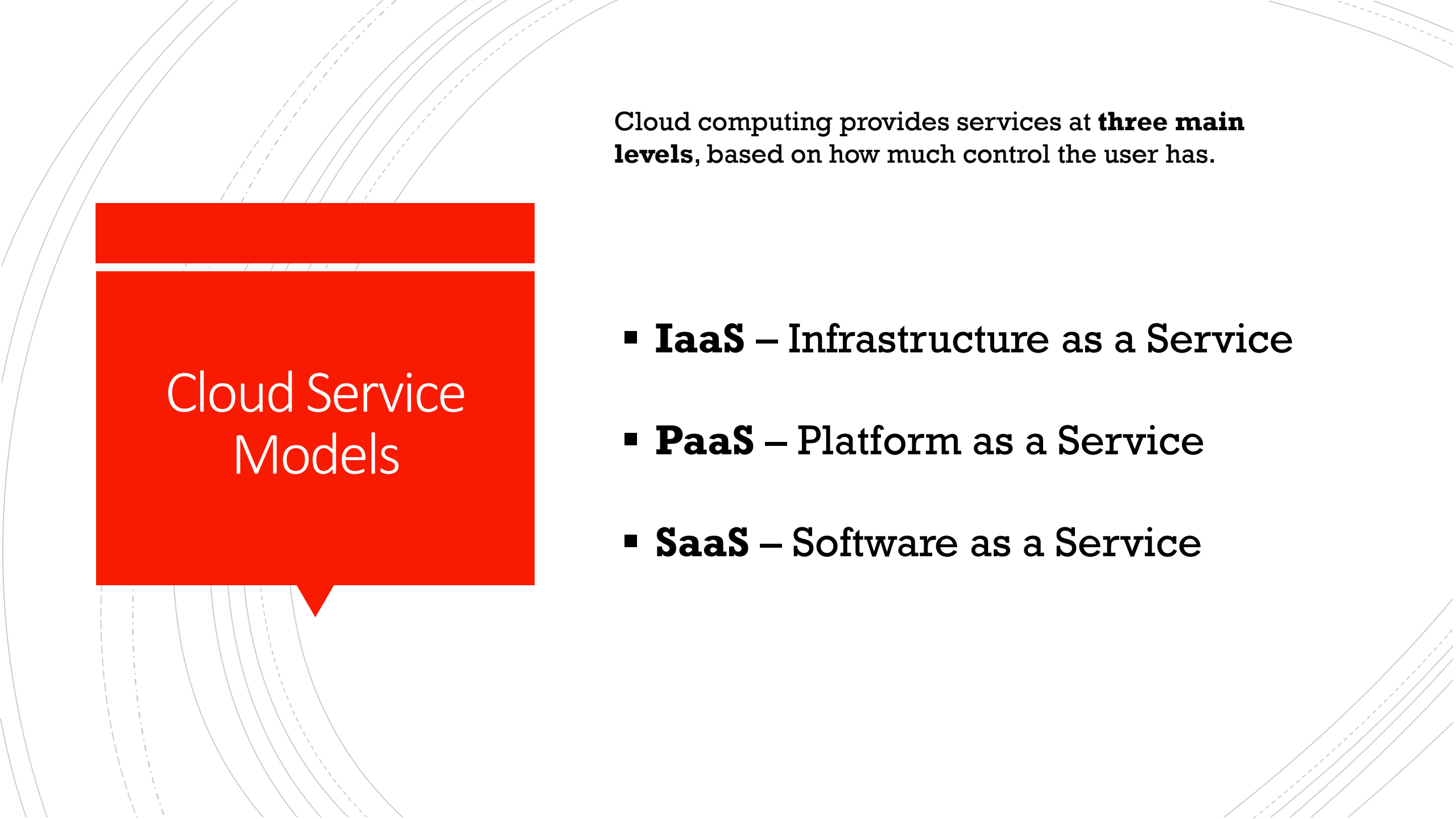
- Kubernetes
- Terraform
- Service Mesh (Istio, Linkerd)
- CI/CD pipelines
- Centralized monitoring tools

Real-World Example

- Large E-commerce Platform
- AWS for backend services
- GCP for analytics
- Azure for identity management

Public vs Private vs Hybrid vs Multi-Cloud

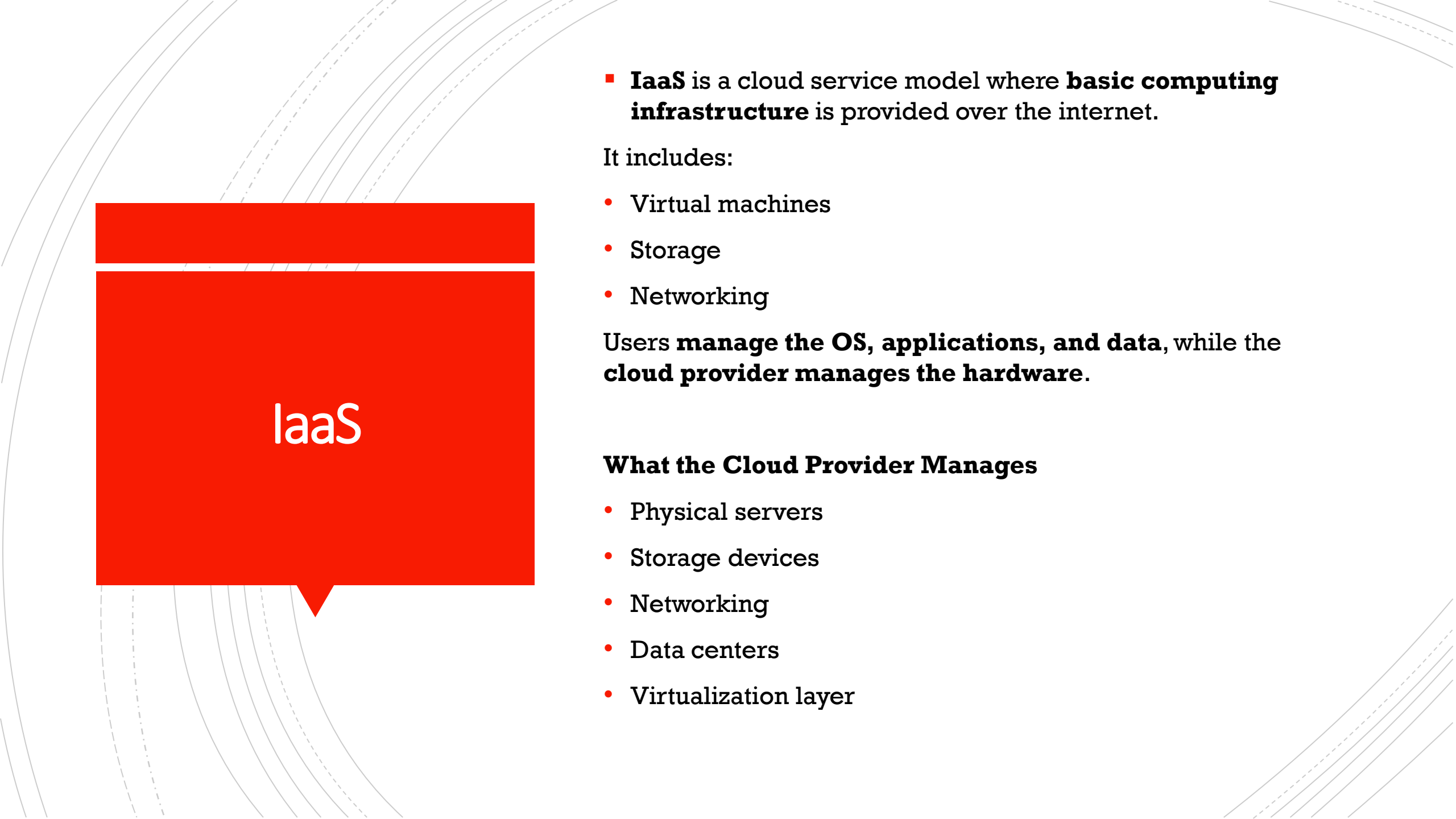
Feature	Public Cloud	Private Cloud	Hybrid Cloud	Multi-Cloud
Definition	Shared cloud services over internet	Dedicated cloud for one organization	Combination of public + private cloud	Use of multiple cloud providers
Infrastructure Ownership	Cloud provider	Organization / Dedicated provider	Shared (public + private)	Multiple cloud providers
Tenancy	Multi-tenant	Single-tenant	Mixed	Mixed
Cost	Low (pay-as-you-go)	High	Medium	Medium-High
Scalability	Very high	Limited	High	Very high
Security	Moderate	Very high	High	High (complex)
Control	Low	Very high	High	Medium
Complexity	Low	Medium	High	High
Vendor Lock-in	High	Low	Medium	Low
Typical Users	Startups, students	Banks, government	Enterprises	Large enterprises
Use Cases	Web apps, SaaS	Sensitive data	Legacy + cloud apps	High availability systems
Examples	AWS, Azure, GCP	OpenStack, VMware	AWS + On-prem	AWS + Azure + GCP

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Cloud Service Models

Cloud computing provides services at **three main levels**, based on how much control the user has.

- **IaaS** – Infrastructure as a Service
- **PaaS** – Platform as a Service
- **SaaS** – Software as a Service



IaaS

- **IaaS** is a cloud service model where **basic computing infrastructure** is provided over the internet.

It includes:

- Virtual machines
- Storage
- Networking

Users **manage the OS, applications, and data**, while the **cloud provider manages the hardware**.

What the Cloud Provider Manages

- Physical servers
- Storage devices
- Networking
- Data centers
- Virtualization layer

What the User Manages

- Operating system
- Runtime & middleware
- Applications
- Data
- Security configurations

Key Characteristics of IaaS

- On-demand resources
- High scalability
- Pay-as-you-go pricing
- Full control over infrastructure
- Suitable for customized setups

Examples of IaaS Providers

- AWS EC2
- Google Compute Engine
- Microsoft Azure Virtual Machines
- Oracle Cloud Infrastructure

Use Cases of IaaS

- Web application hosting
- Development and testing environments
- Disaster recovery
- Big data processing
- Running legacy applications

Advantages of IaaS

- No hardware investment
- Flexible and scalable
- Full control over servers
- Suitable for DevOps and CI/CD

Disadvantages of IaaS

- User responsible for OS and security
- Requires technical expertise
- More management compared to PaaS/SaaS

IaaS vs On-Premise

IaaS

Pay-as-you-go

Highly scalable

Fast provisioning

Provider managed hardware

On-Premise

High upfront cost

Limited scalability

Slow setup

Self managed hardware

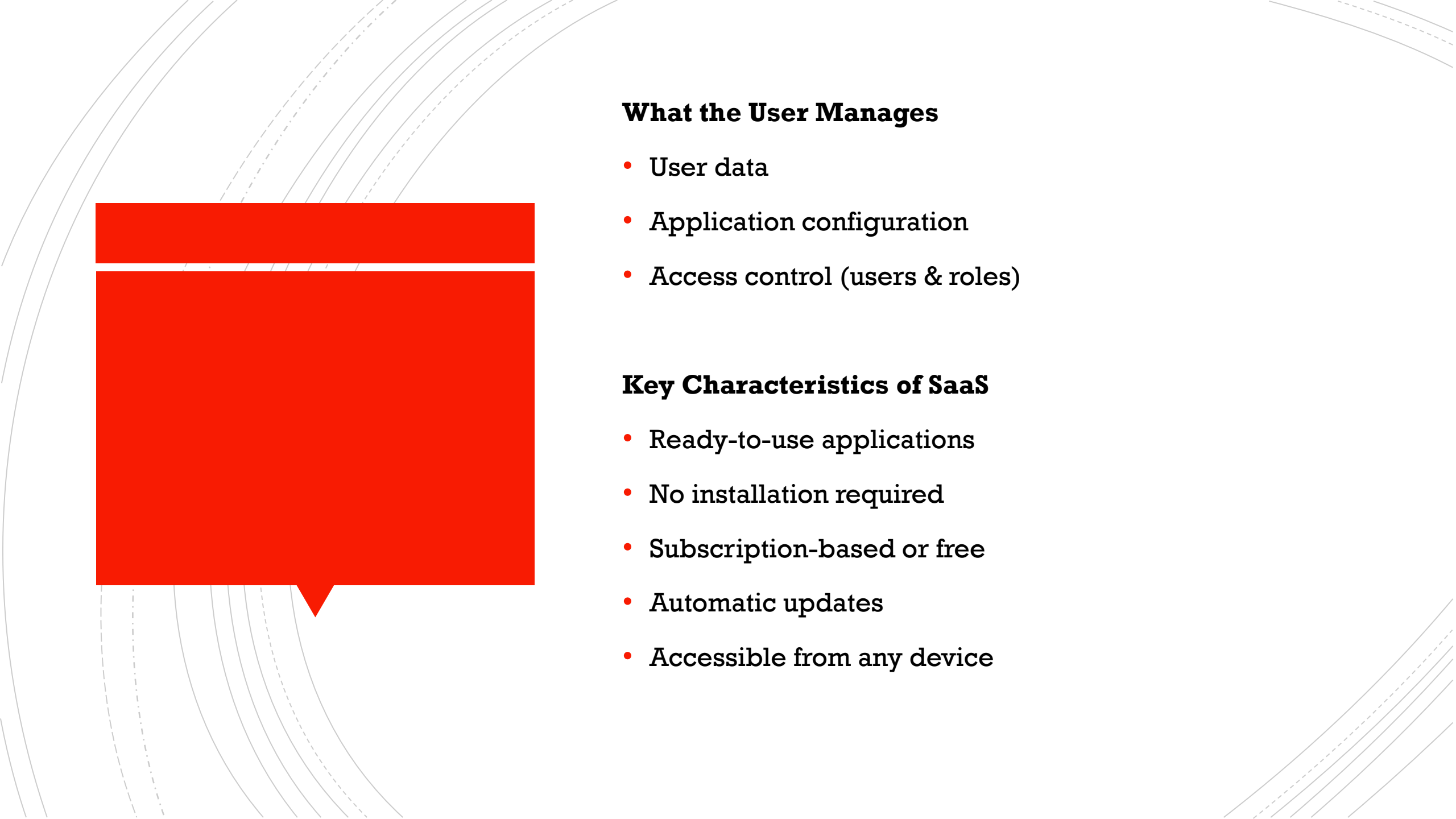


SaaS

- **SaaS** is a cloud service model where **complete software applications** are delivered over the internet.
- Users **access the software via a web browser**, and **do not manage** servers, operating systems, or application updates.

What the Cloud Provider Manages

- Infrastructure (servers, storage, networking)
- Operating system
- Application software
- Security patches & updates
- Availability and scalability

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What the User Manages

- User data
- Application configuration
- Access control (users & roles)

Key Characteristics of SaaS

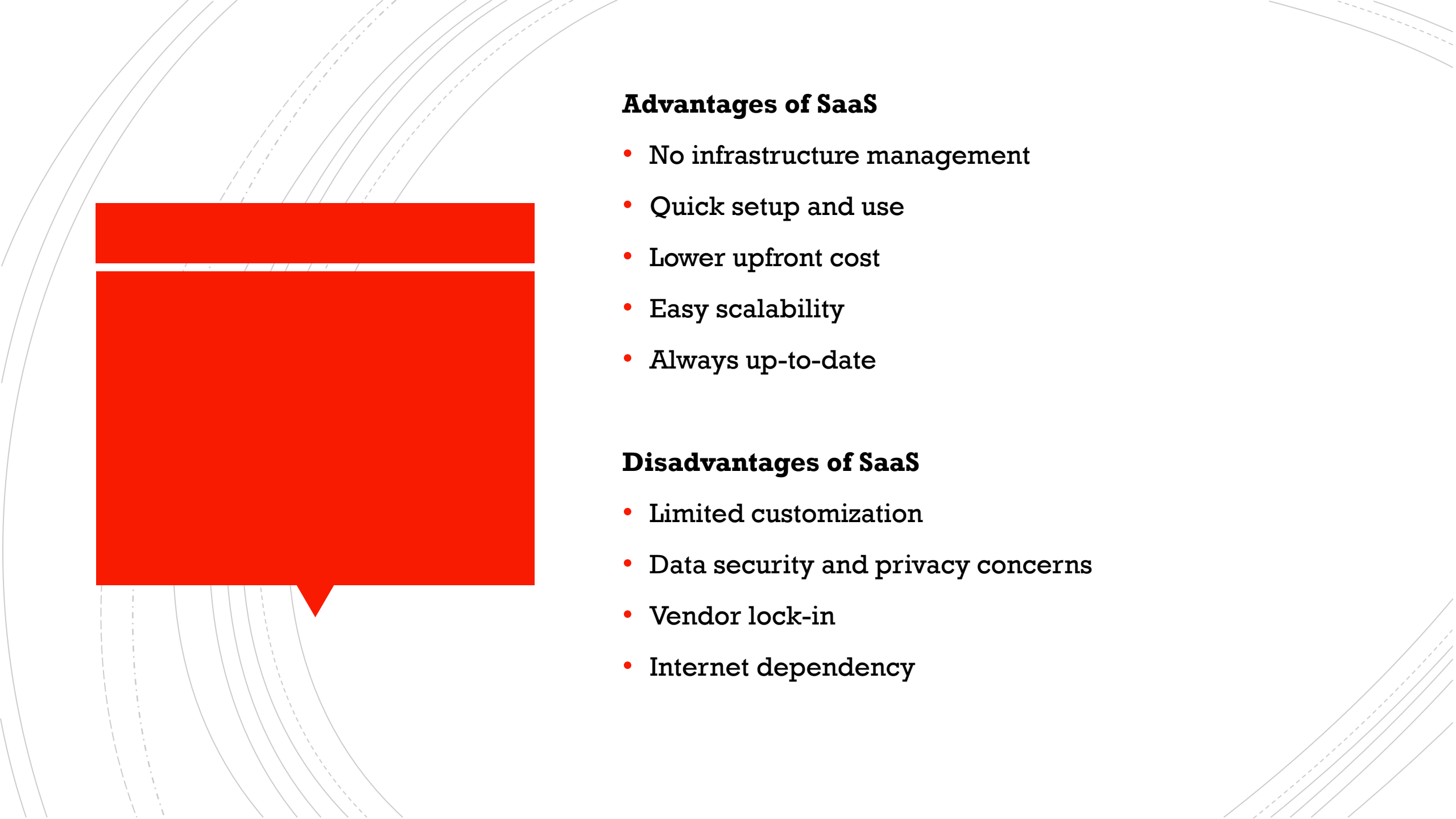
- Ready-to-use applications
- No installation required
- Subscription-based or free
- Automatic updates
- Accessible from any device

Examples of SaaS Applications

- **Google Workspace (Gmail, Docs)**
- **Microsoft 365**
- **Salesforce**
- **Zoom**
- **Dropbox**

Use Cases of SaaS

- Email and communication
- Customer relationship management (CRM)
- Online collaboration
- E-learning platforms
- Accounting and finance software

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Advantages of SaaS

- No infrastructure management
- Quick setup and use
- Lower upfront cost
- Easy scalability
- Always up-to-date

Disadvantages of SaaS

- Limited customization
- Data security and privacy concerns
- Vendor lock-in
- Internet dependency

SaaS vs Traditional Software

SaaS

Cloud-based

Subscription model

Auto updates

Accessible anywhere

Traditional Software

Installed locally

One-time license

Manual updates

Device dependent



PaaS

What Is PaaS?

- **PaaS** is a cloud service model that provides a **complete platform for application development, testing, and deployment**.
- Developers focus on **writing code**, while the cloud provider manages the **infrastructure and runtime environment**.

What the Cloud Provider Manages

- Servers, storage, networking
- Operating system
- Runtime environment
- Middleware
- Scaling & availability



What the User Manages

- Application code
- Application configuration
- Data

Key Characteristics of PaaS

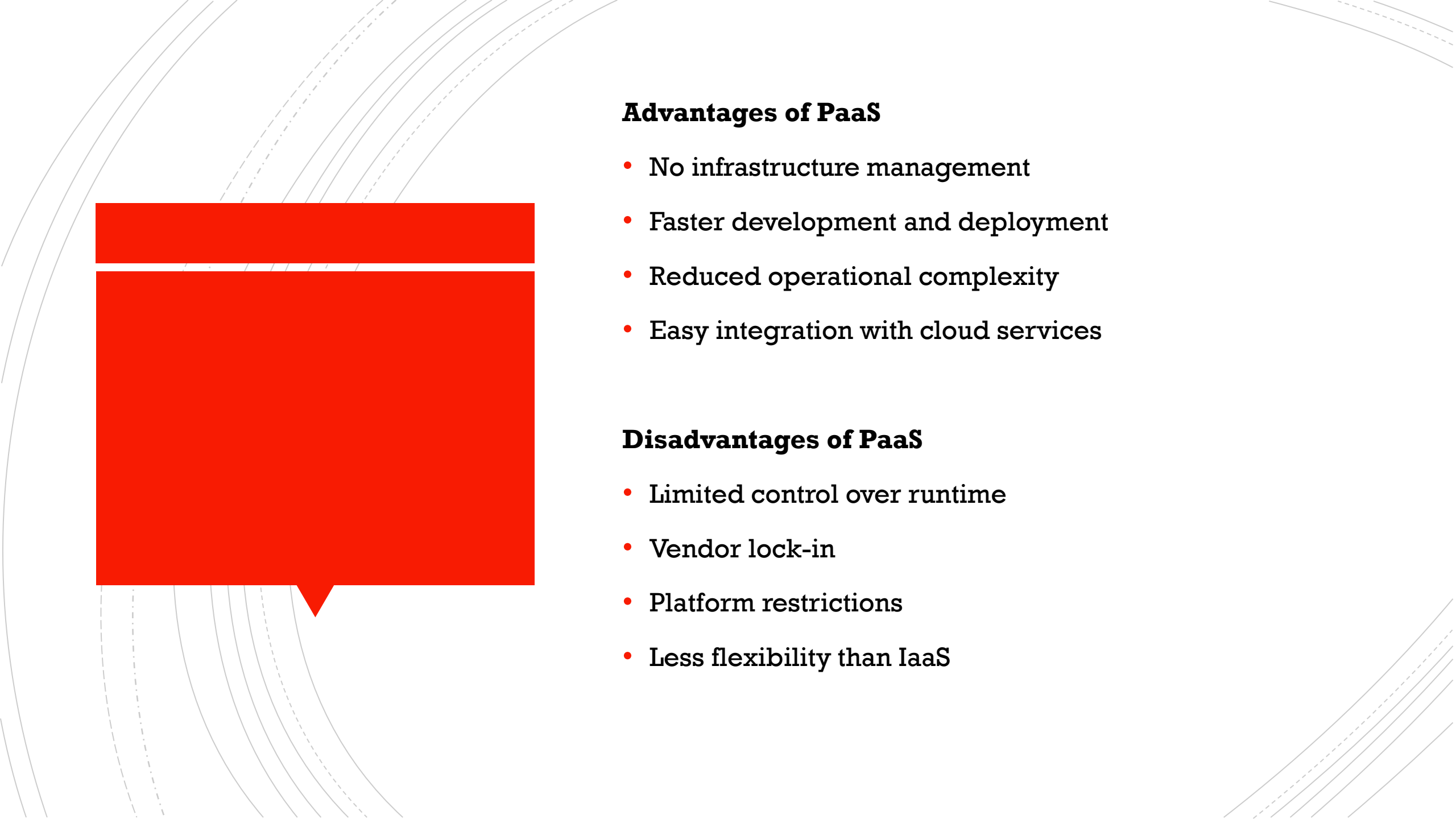
- Developer-friendly
- Faster application development
- Built-in scalability
- Integrated development tools
- Supports CI/CD pipelines

Examples of PaaS Providers

- **Google App Engine**
- **AWS Elastic Beanstalk**
- **Microsoft Azure App Service**
- **Heroku**

Use Cases of PaaS

- Web application development
- API development
- Microservices applications
- Agile and DevOps workflows
- Student and startup projects



Advantages of PaaS

- No infrastructure management
- Faster development and deployment
- Reduced operational complexity
- Easy integration with cloud services

Disadvantages of PaaS

- Limited control over runtime
- Vendor lock-in
- Platform restrictions
- Less flexibility than IaaS

PaaS vs IaaS vs SaaS

Feature	IaaS	PaaS	SaaS
User manages	OS & apps	Apps & data	Only data
Flexibility	High	Medium	Low
Ease of use	Low	Medium	High
Target users	System admins	Developers	End users

Evolution of Cloud Computing

- Pre-Cloud Era
- Virtualization Era (2000–2005)
- Birth of Cloud Computing (2006)
- Growth Era (2010–2020)
- Modern Cloud (2020–Present)

<https://www.geeksforgeeks.org/cloud-computing/evolution-of-cloud-computing/>

Cloud Advantages

- Benefits of Cloud Computing

- Trade fixed expense for variable expense
- Benefit from massive economies of scale
- Stop guessing capacity
- Increase speed and agility
- Stop spending money running and maintaining data centers
- Go global in minutes

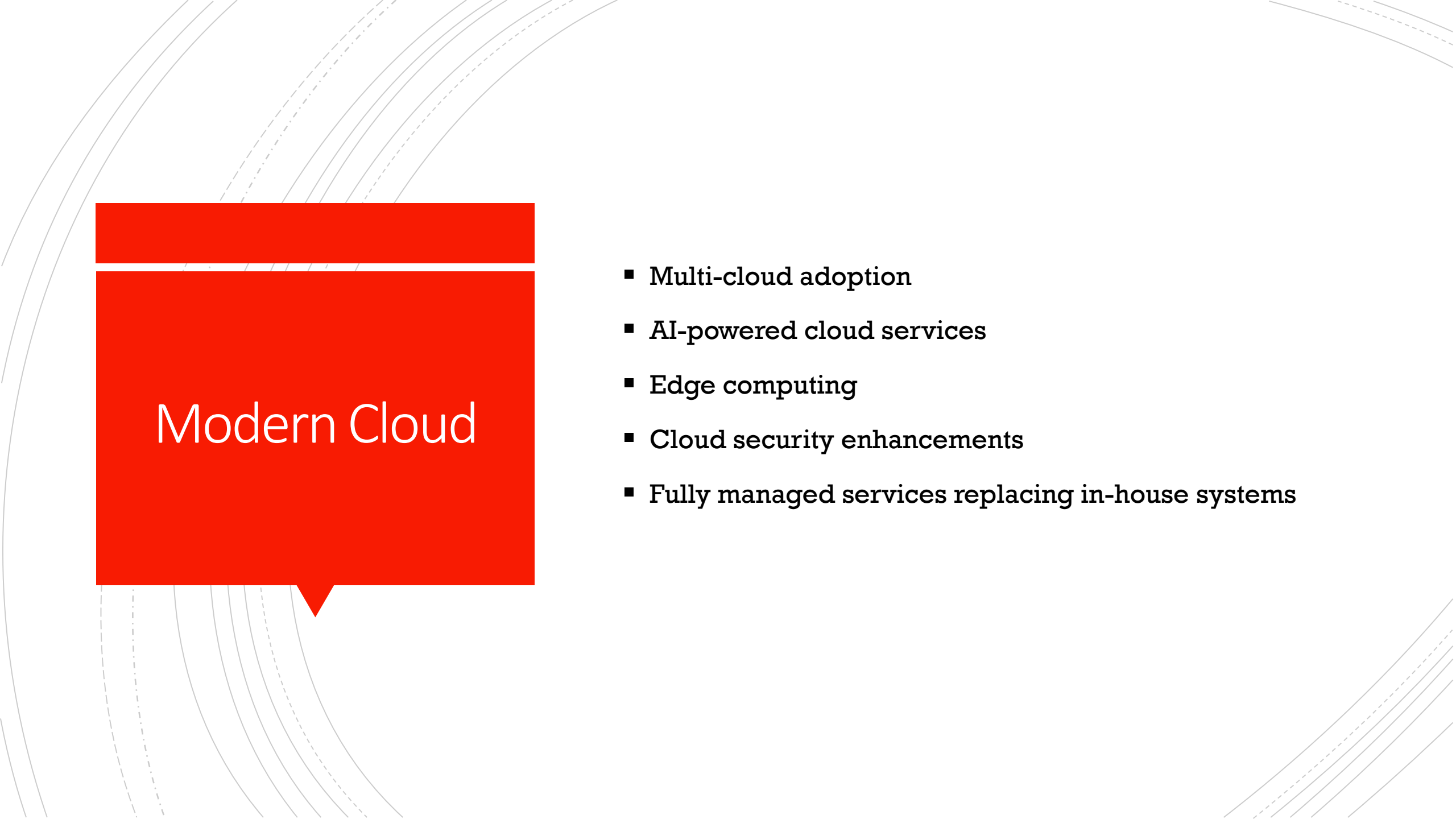
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Cloud Challenges

- **Data Security and Privacy**
- **Cost Management**
- **Multi-Cloud Environments**
- **Performance Challenges**
- **Interoperability and Flexibility**
- **High Dependence on Network**
- **Lack of Knowledge and Expertise**

Use Cases of Cloud

- Web & Mobile Application Hosting
- Data Storage & Backup
- Big Data Processing & Analytics
- Machine Learning & Artificial Intelligence
- Software Development & Testing
- Internet of Things (IoT)
- Disaster Recovery & Business Continuity
- Streaming & Content Delivery
- Enterprise Applications
- Education & E-Learning
- Gaming
- Government & Smart Cities

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Modern Cloud

- **Multi-cloud adoption**
- **AI-powered cloud services**
- **Edge computing**
- **Cloud security enhancements**
- **Fully managed services replacing in-house systems**

Future of Cloud

- Quantum cloud computing
- Increased automation
- AI-driven infrastructure
- More edge computing
- Fully serverless systems
- Future is cloud-native + AI-powered



Q & A?